

TrES-2b

R-2520

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_100930 · created 2026-08-25T01:06:21+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2455469.65091 +/- 3.6e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1364 +/- 0.0091
Transit depth (Rp/Rs)^2	1.86 +/- 0.25 [%]
Orbital Inclination (inc)	85.07 +/- 0.72
Airmass coefficient 1 (a1)	1.153 +/- 0.0092
Airmass coefficient 2 (a2)	-0.111 +/- 0.012
Scatter in the residuals of the lightcurve fit is	0.79 %
Best Comparison Star	None
Optimal Aperture	3.34
Optimal Annulus	6.25
Transit Duration (day)	0.0904 +/- 0.0074

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1364 ± 0.0091 vs 0.1254 (deviation 1.21σ)
Duration fit vs published	0.0904 d vs 0.0746 d (deviation 2.14σ)
Mid-time O–C	0.0 min
Depth SNR	15.0
Residual scatter	0.79 %

Light curve

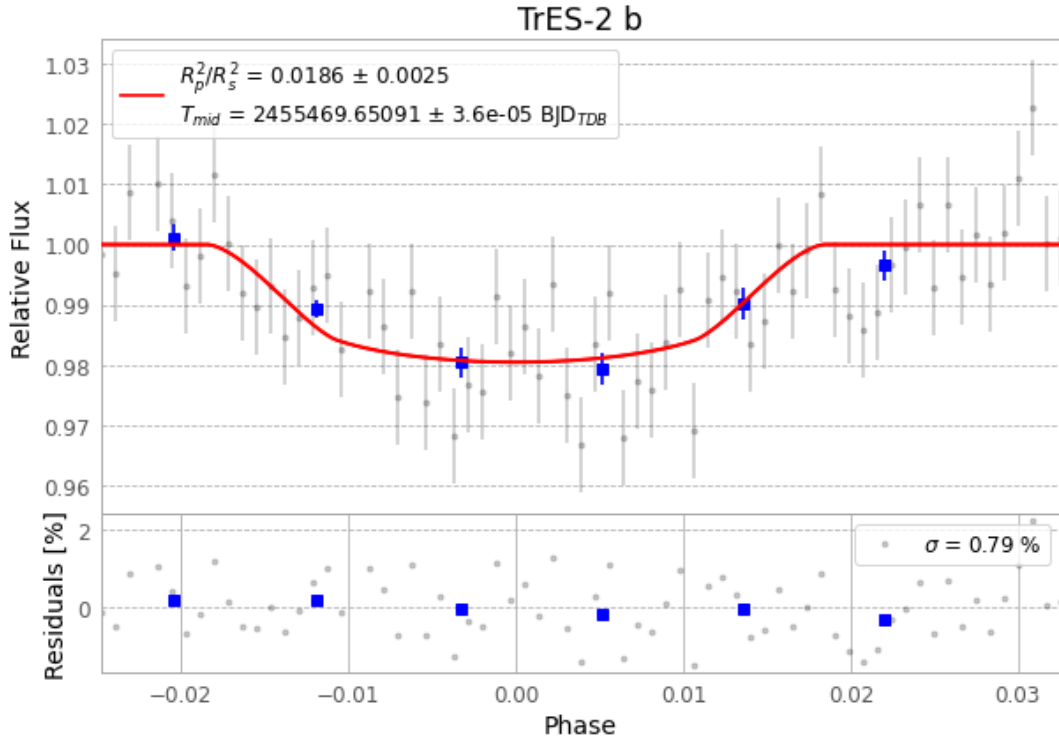


Figure 1 — FinalLightCurve_TrES-2 b_2010-09-29.png (SHA-256 bae9f2c08612dd5c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [269, 202], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4653db54fc2185ba...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

69 FITS frames (45 MB), TRES-2100930020614.FITS → TRES-2100930053014.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-100930.log (94334e04fbce...), FinalLightCurve_TrES-2 b_2010-09-29.png (bae9f2c08612...), FinalLightCurve_TrES-2 b_2010-09-29.csv (05a1c6ee41a7...)

Compute resources

Wall time 113.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-25 01:06Z.